Intelligent retail & supply chain planning

Implemented architecture, quantitative decision logic and validation evidence | repository documentation reviewed 17 October 2026

System topology

Boundary	Implemented responsibility
React / TypeScript / Vite	Inventory, forecasting, scenario, recommendation and reporting workspaces. Typed Fetch API.
FastAPI + PostgreSQL 17	JWT/RBAC, domain transactions, inventory locks, review/approval, persisted jobs and append-oriented audit.
Serial job worker + outbox	Forecast/risk/optimisation pipeline; transactional event capture, retry state and warehouse refresh.
scikit-learn + OR-Tools	Per-SKU/store Ridge demand forecasts; CP-SAT joint transfer and MOQ procurement allocation.
DuckDB / provider boundaries	Independent analytical fact/dimension projections; optional GCP SDK adapters.

Forecasting contract

Component	Technical implementation
Estimator	StandardScaler + Ridge(alpha=20); separate fit per product/store daily sales series.
Features	Lags 1/7/14, rolling means 7/28, weekday and annual sine/cosine, elapsed trend, promotion flag.
Horizons / uncertainty	7, 14, 30 days; recursive multi-step predictions; empirical daily 90% intervals from chronological calibration.
Evaluation	MAE, RMSE, WAPE, nonzero MAPE, bias and coverage; untouched test window and later observed-outcome evaluation.

Optimisation contract

OR-Tools CP-SAT jointly chooses integer stock transfers, binary route activation and supplier MOQ batches. The objective combines transfer/procurement cost, holding-cost credit and unmet-demand penalties; location safety, inbound capacity, lead times, source surplus, minimum transfer size and procurement holds remain hard constraints.

Decision variable	Guard / purpose
Transfer units x(s,d,p)	Bounded by source surplus and destination deficit; a route requires at least 10 units.
Route activation z	Adds fixed route activation cost to the objective.
Supplier batches b	Integer MOQ, supply availability, lead-time eligibility and hold restrictions.
Unmet units u	Explicitly penalised rather than silently assumed fulfilled.

Operational controls & verification

Persistence and review gates, measured test evidence, and cloud deployment status

Workflow and transactional integrity

Control	Contract / testable effect
Recommendation state	GENERATED -> UNDER_REVIEW -> APPROVED / REJECTED -> EXECUTED -> MEASURED.
Execution gate	Only operations/admin roles can approve or execute; changed stock/capacity is revalidated and duplicate execution rejected.
Atomic operational state	Sales and movements lock affected inventory and commit state, audit and outbox events together.
Scenario isolation	Copied input state persists scenario outcomes without mutating operational inventory.
Job persistence	QUEUED/RUNNING/COMPLETED/FAILED jobs with progress and failure details; local worker uses PostgreSQL advisory locking.
Audit	PostgreSQL trigger prevents update/delete of append-oriented audit records (database administrators retain higher privilege).

Verified local evidence - 1 October 2026

Check / fixture	Recorded result
Seed scope	18 products; 4 stores; 2 distribution centres; 5 suppliers; 108 stock positions; 39,456 daily sales observations.
Forecast check	One rice/Johannesburg series: 7d WAPE 12.27%; 14d 11.36%; 30d 14.24% (not fleet-wide accuracy).
Recorded allocation	CP-SAT OPTIMAL: 110 Cape Town + 73 Pretoria + 25 Johannesburg DC units to Johannesburg. Estimated transfer cost R610.60.
Automated verification	24 pytest passed; Python/TS lint passed; React production build passed; smoke PASS; 2 Playwright tests passed.

API surface (selected)

POST /api/forecasts/jobs -> HTTP 202 pipeline job; GET /api/forecasts; POST /api/scenarios -> isolated planning job; POST /api/recommendations/{id}/actions -> review/approve/reject/execute/measure; POST /api/sales -> locked stock decrement; POST /api/purchase-orders/{id}/receive -> simulated receiving; GET /api/analytics/{report}.

GCP integration status

SDK adapters implemented and locally contract-tested: BigQuery load/query, Pub/Sub publish, Cloud Storage upload, Vertex endpoint prediction and Gemini evidence explanation. **Terraform prepared but not applied:** Cloud SQL, Cloud Run API/web, migration job, BigQuery, Pub/Sub, GCS, IAM, Secret Manager, Artifact Registry and monitoring. No live GCP calls or production deployment are claimed. Local worker execution is serial, and modelled savings are not realised returns.